

GAUSFER 3D

MINI PROJECT REPORT

Submitted by

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*in partial fulfillment for the award of the degree
of*

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

Under the guidance of

SNEHA S

Assistant professor



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING AND MANAGEMENT PUNNAPRA**

(Under CAPE, Estd by Govt. of Kerala)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING AND MANAGEMENT PUNNAPRA

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CERTIFICATE

This is to certify that the report document entitled “**Gausfer 3D**” submitted by **Amal S Kumar (PRP23CS007)**, **Aleesha M (PRP23CS006)**, **Ambili Ambaran S (PRP23CS009)**, **Aswin S Tushar (PRP23CS024)** to the APJ Abdul Kalam Technological University in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology** in **Computer Science and Engineering** is a bonafide record of the project work carried out by them under our supervision.

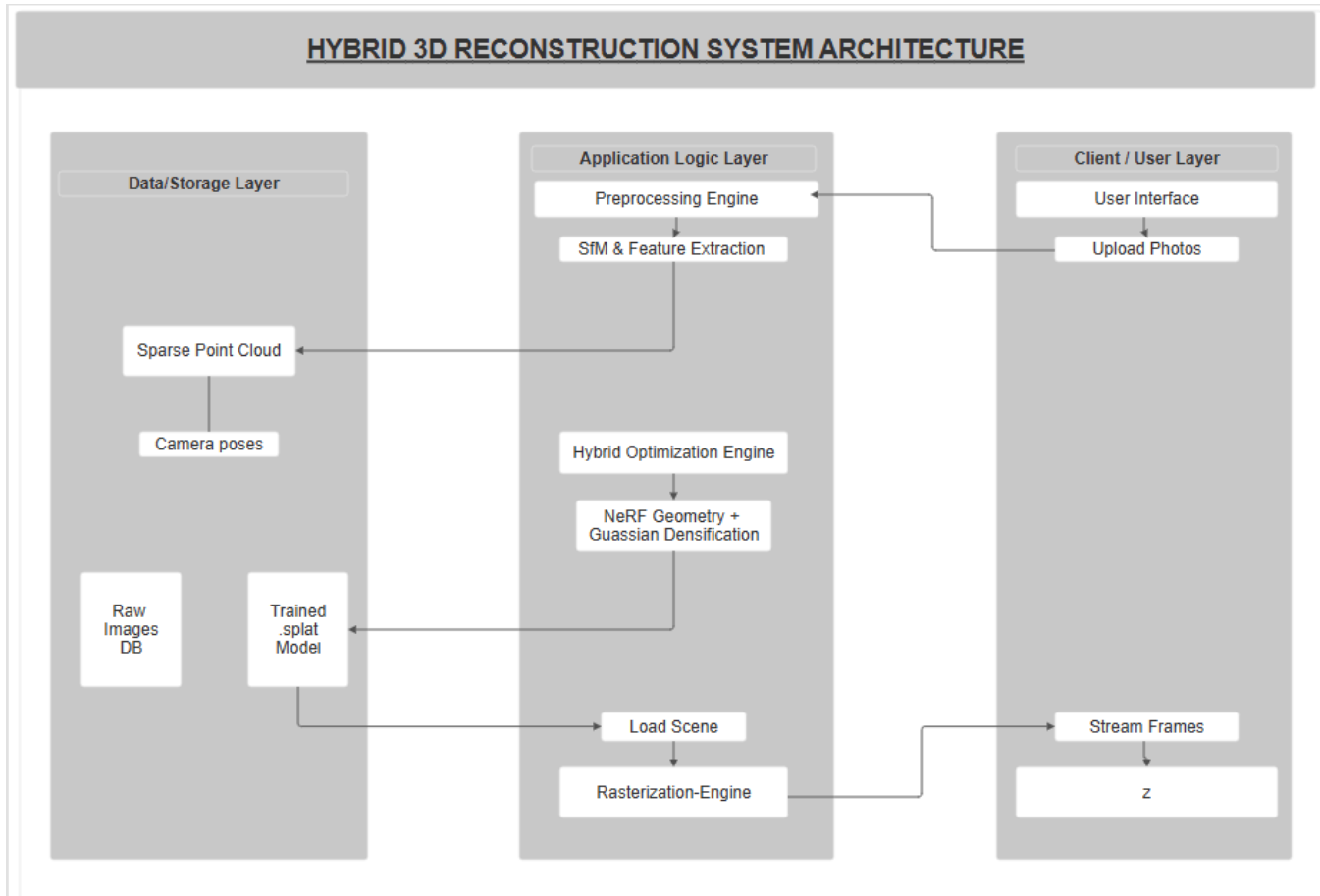
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5.1.2.1 ARCHITECTURAL DESIGN



The architecture follows a three-layer model:

1. **Data Storage Layer:** Manages sparse point clouds and camera poses derived from COLMAP.
2. **Application Logic Layer:** Contains the Pre-processing Engine (Laplacian filtering), the Hybrid Optimization Engine (NeRF + Gaussian), and the Rasterization Engine.
3. **Client User Layer:** Provides the interface for video/image uploads and real-time frame streaming (>100 FPS).

5.1.3 DATA DESIGN

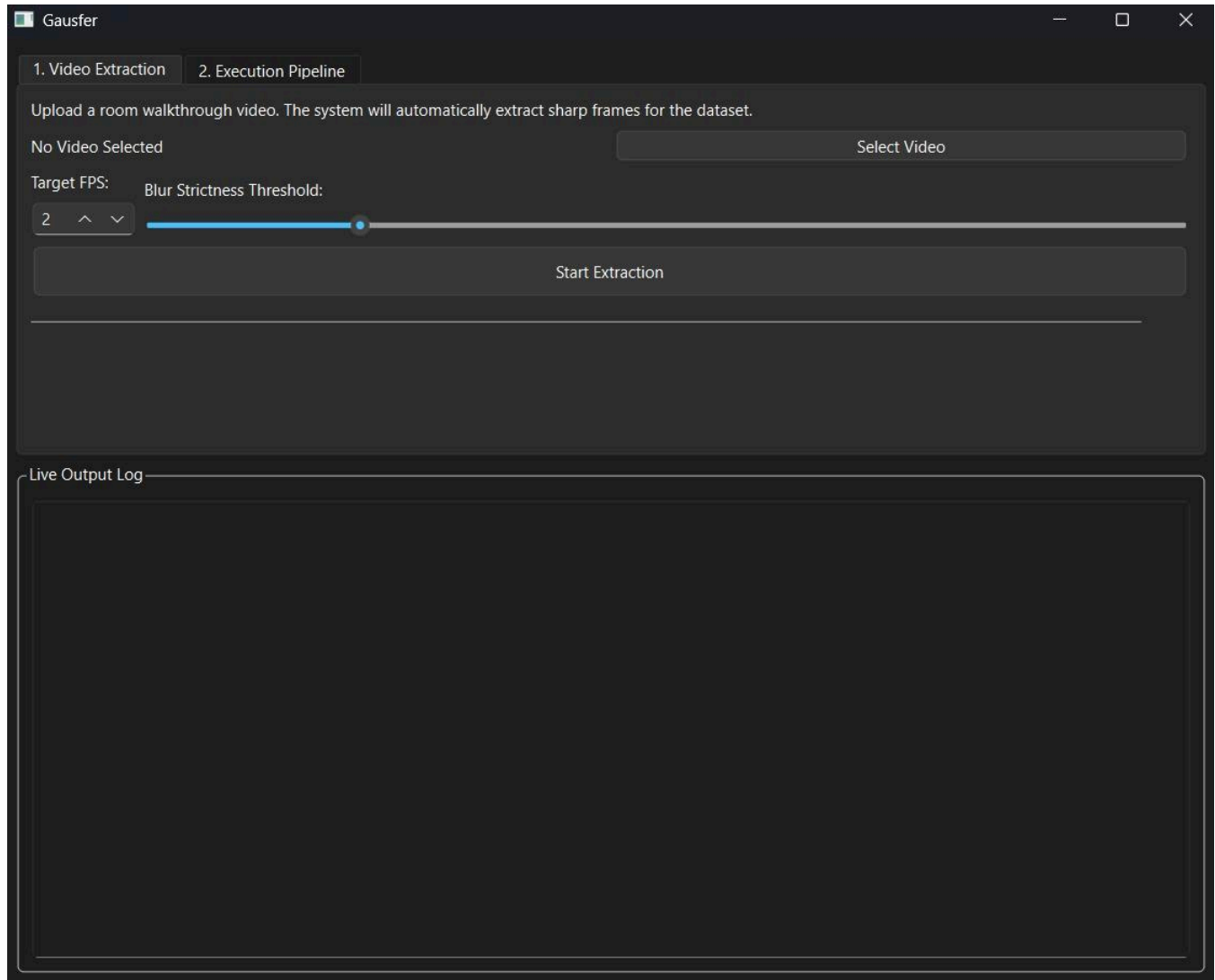
The data design organizes the transition from 2D pixel data to 3D spatial representations, optimizing storage for GPU-accelerated rendering..

a. DATA DESCRIPTION

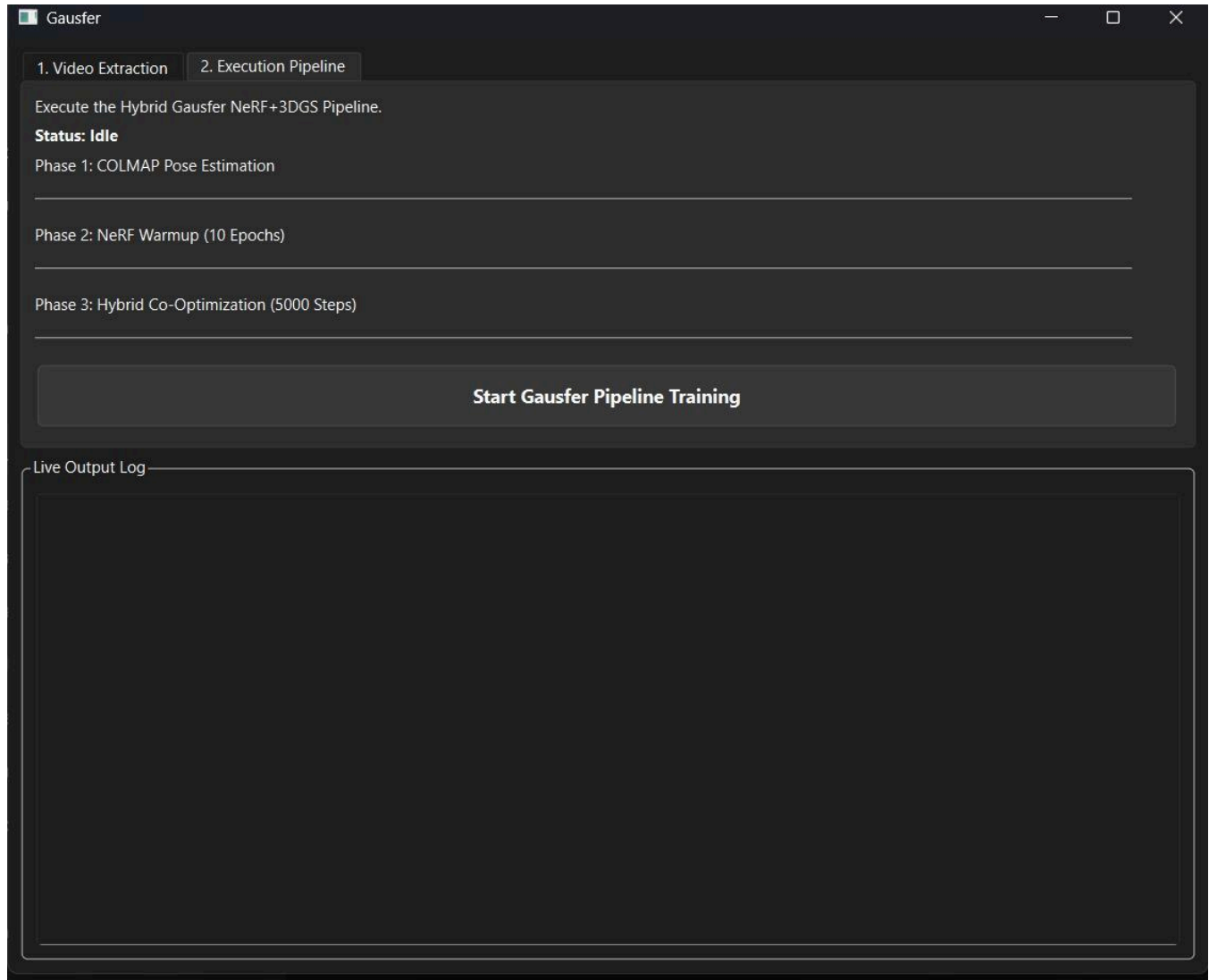
Data includes raw 2D video frames, extracted camera intrinsics/extrinsics, voxel grids (256^3), and explicit Gaussian primitives (position, opacity, and gradients)

b. DATA DICTIONARY

σ (Sigma): Volumetric density value used to identify solid surfaces.



GAUSFER 3D: FROM FLAT PIXELS TO IMMERSIVE REALITY



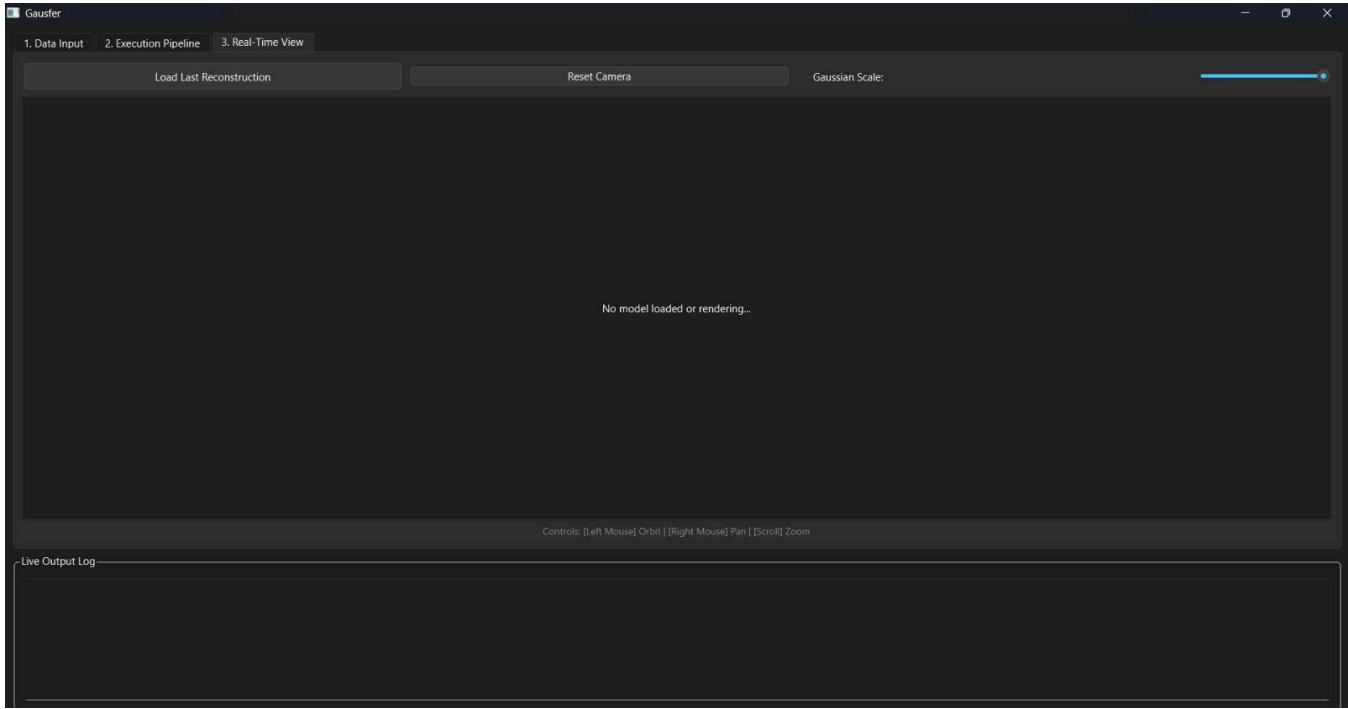


Fig: User Interface

5.1.4 REQUIREMENT TRACABILITY MATRIX

Requirement ID	Description	Test Case	Tester Notes / Evidence
REQ-01	Automated Data Filtering: System must automatically discard motion-blurred frames	Upload video with intentional high-speed camera movement	Uses a Laplacian variance threshold ($\sigma^2_{\text{Laplacian}} > 100.0$) to filter out blurry imagery.
REQ-02	Pose Estimation: System must solve for camera intrinsics and extrinsics from 2D frames	Run pipeline on image set with diverse viewpoints.	Utilizes a CUDA-accelerated COLMAP wrapper to extract 3x 3 rotation matrices (R)
REQ-03	Neural Scaffolding: System must establish a	Execute 10-epoch warmup period for the	Generates precise volumetric point cloud seeds; achieves

CHAPTER 8

RESULTS

8.1 GENERAL

Experimental evaluation demonstrates that the proposed hybrid system effectively reconstructs 3D environments from 2D video imagery and optimizes novel view synthesis with high performance. The NeRF-based initialization module generates precise volumetric point cloud seeds, achieving a rapid geometric convergence within just 20 epochs of NeRF and 10,000 steps of gaussian splatting, yielding an overall Peak Signal-to-Noise Ratio (PSNR) of 20.29 dB, a Structural Similarity Index (SSIM) of 0.7150, and an LPIPS score of 0.18. For spatial optimization, the Hybrid Co-Optimizer classifies rendering regions into high, medium, and low density clusters while converging in merely 5,000 steps, demonstrating strong capability in capturing the relationship between initial NeRF volumetric density and the final 3D Gaussian distribution. By enforcing a dynamic NeRF density threshold and an opacity limit and a hard limit of 1 Million Splats (downsized after the steps) the system efficiently restricts the computational footprint to an 8 GB VRAM budget. The integration of NeRF-guided initialization and 3DGS dynamic pruning significantly enhances the overall rendering capability, as the system not only identifies correct structural foundations but also predicts and populates high-frequency textural details in real-time (>20 FPS). Furthermore, the inclusion of an automated Laplacian variance preprocessing layer improves system usability by filtering out blurry imagery—environmental factors that most strongly influence geometry degradation—thereby increasing structural transparency and data-extraction reliability.



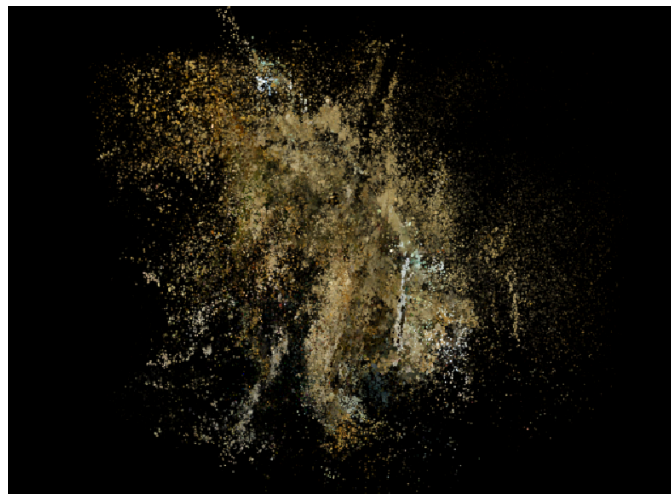
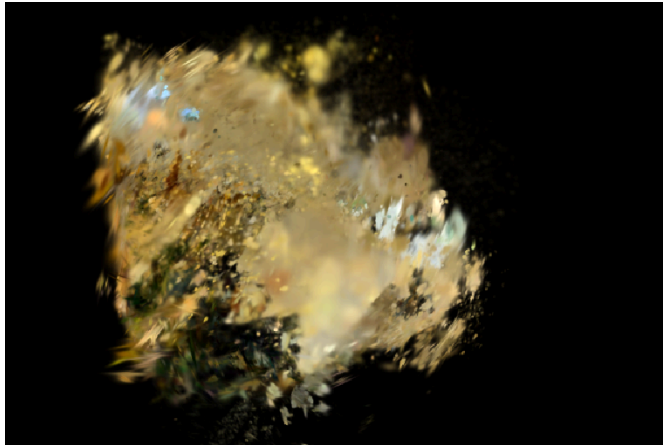


FIGURE OBTAINED FROM TRAINING
(TRAINING ISSUES DUE TO HARDWARE AND TRAINING CONSTRAINTS)